# Visualization of Single Layer Perceptron



$$\boldsymbol{d}_1(\boldsymbol{X}) = \underline{W_1}^T \boldsymbol{X} = w_{11}x_1 + w_{12}x_2 + \ldots + w_{1n}x_n + b_1 * 1$$

$$\boldsymbol{d}_2(\boldsymbol{X}) = \underline{W_2}^T \boldsymbol{X} = w_{21}x_1 + w_{22}x_2 + \ldots + w_{2n}x_n + b_2 * 1$$

$$\boldsymbol{d}_m(\boldsymbol{X}) = \underline{W_m}^T \boldsymbol{X} = w_{m1}x_1 + w_{m2}x_2 + \ldots + w_{mn}x_n + b_m * 1$$

***Multi-Class Classification***

